

Datasheet

Technical Datasheet

Virtual Battery Factory · case t7_r1 · 2026-08-25

Technical Datasheet — HEV Fast-Charge Pouch Cell (Case t7_r1)

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Field	Value	Source
Rated capacity	6.03 Ah (nominal 6.0313 Ah; simulated 6.032209 Ah, DFN)	params_v11.json; cell/r5_v11_1c_dfn.json:capacity_ah
Nominal voltage / voltage window	3.72 V (discharge midpoint) / 2.5 – 4.2 V	cell/r5_v11_energy.json:midpoint_voltage_v; parameter set
Rated energy	21.797 Wh (1C discharge integration)	cell/r5_v11_energy.json:energy_wh
Gravimetric energy density	483.25 Wh/kg (contract formula, electrolyte excluded)	cell/r5_v11_energy.json:energy_density_wh_kg
Volumetric energy density	920.38 Wh/L	cell/r5_v11_energy.json:energy_density_wh_l
Maximum continuous discharge rate	1C (6.03 A) — simulated at 25 °C ambient; higher rates not adjudicated	cell/r5_v11_1c_dfn.json
Fast-charge capability	4C (24.1 A) at 45 °C ambient: no lithium plating (anode surface min +0.0476 V); max cell temperature 330.13 K (56.98 °C)	cell/r5_v11_4c45.json
DC internal resistance (DCR)	2.882 mΩ	cell/r5_v11_energy.json:dcr_ohm
Power density	32 559.7 W/kg	cell/r5_v11_energy.json:power_density_w_kg
Operating temperature range	Verified by simulation at 298.15 K (1C discharge) and 318.15 K (4C charge, 100-cycle aging). Other temperatures: not simulated.	run outputs (ambient per protocol)
Cycle life (SEI)	SEI thickness 465.47 nm after 100 cycles at 45 °C (≤ 550 nm limit). Cycle count to failure beyond 100 cycles: not simulated (aging model runs 100 cycles only).	cell/r5_v11_aging45.json:sei_thickness_nm_end
Cycle-life capacity trajectory	Per-cycle capacities show a model artifact (thick SEI film resistance dominates at μm-scale film); the adjudicable metric is sei_thickness_nm_end — see DVPR note.	cell/r5_v11_aging45.json; evaluate R3–R5 notes
Safety — fast charge	No plating at 4C/45 °C (anode potential never below 0 V)	cell/r5_v11_4c45.json
Safety — nail penetration	10 W short-circuit heat source, virtual TR test: thermal runaway NOT triggered (T_max 336.32 K = 63.17 °C; dT/dt max 0.245 K/s), with h = 50 W/(m²·K) cooling (hA = 0.2655 W/K). Cooling floor: h = 40 W/(m²·K) (h=30 triggers at 666 s per sensitivity run r5_v8_nail_h30). Physical nail test: N/A (requires experiment).	cell/r5_v11_nail.json; cell/r5_v8_nail_h30.json
Dimensions	Electrode stack: 65 mm × 1580 mm × 0.2306 mm (parameter-set pouch geometry; shell not modeled)	parameter set; cell/r5_v11_energy.json:thickness_m
Mass	45.11 g cell stack (contract, electrolyte excluded); +8.92 g electrolyte (literature estimate) ≈ 54.03 g total	cell/r5_v11_energy.json:mass_kg; process formula
Charge voltage limit	4.2 V (parameter set)	parameter set

Notes: all values are virtual — simulated with PyBaMM (DFN for rated data) and the three-side-reaction thermal-runaway ODE; no physical cell was built or tested. True first-principles (DFT/MD) endorsement skipped: real_compute = false (entry 0).